

A Hering prior in the gaze-holding integrator bounds adaptive correction of strabismus

Draft manuscript

Abstract

Comitant strabismus is conventionally framed as either a peripheral plant abnormality or a central deficit of binocular fusion. We argue for a third framing, complementary to both: that a substantial component of the residual binocular misalignment that survives long-standing adaptation is a signature of structural priors in the recurrent connectivity of the binocular neural integrator. The integrator's continuous-attractor architecture imposes a specific manifold of held postural states on motoneuron activity space, and this manifold geometry, rather than any feedforward yoking rule, is the operative structural prior on binocular alignment. Under symmetric plant geometry, the natural manifold structure — a low-dimensional sheet decomposing into conjugate and vergence directions — is exactly what binocular alignment requires. Under any asymmetric plant the alignment-correct manifold differs from the natural one, and reshaping the integrator's connectivity to track it incurs both a homeostatic regularization cost and a stability cost. We formalize the framework as a constrained optimization on the integrator's recurrent weight matrix, and identify the predicted residuals analytically for the linearized regime: persistent misalignment coupled with subclinical gaze-holding leak. We then show that the linear analysis has a representational ceiling: nonlinear plant features such as muscle saturation produce a curved alignment-correct manifold that no flat-kernel integrator can contain, and the resulting gaze-dependent residual is the formal version of incomitant strabismus. The compensation strategy available for nonlinear residuals — gaze-dependent cerebellar gain-field modulation — has its own representational and learnability constraints, and the system's choice between flat-kernel approximation and gain-field engagement explains spread of comitance as a cost-benefit optimum. We ground the abstract priors in brainstem anatomy and discuss the relationship of the framework to the structurally nonlinear control machinery characterized in the Crawford and Tweed literatures, arguing that adaptive plasticity operates in an approximately linear regime even though the larger oculomotor system is demonstrably nonlinear.

1. Introduction

The binocular oculomotor plant is a 12-input, 6-output device, and its biological controller has access to vastly more degrees of freedom than the alignment problem strictly requires. Yet binocular misalignment is common, and adaptation often fails to fully correct it even given decades of consistent error feedback. The mismatch between abundant peripheral controllability and incomplete adaptive correction is the puzzle this paper is concerned with.

The traditional framing of this puzzle invokes Hering's law of equal innervation, the proposition that the brain issues a unified conjugate command plus a unified vergence command, and that this yoked architecture limits what adaptation can install. A century of clinical thinking, and the standard interpretation of the Hess screen, builds on this premise. But the strict-Hering account has been steadily eroded by experimental work showing that the system has more than two adaptive degrees of freedom: dichoptic saccade adaptation produces opposite-direction corrections in the two eyes, premotor neurons in the paramedian pontine reticular formation encode monocular rather than purely conjugate commands, and saccade-vergence interactions reveal mixed rather than purely additive control. If the brain were strictly Hering, none of this should occur.

We propose that the structural prior responsible for residual misalignment lives at a different level of the system entirely: not in a feedforward yoking rule on the command pathway, but in the recurrent dynamics of the neural integrator. The integrator is the postural controller. Its connectivity defines the manifold of stable held states the system can occupy, and this manifold has a specific low-dimensional geometry that emerges from the requirement of accurate gaze-holding under metabolic and stability constraints. That geometry is the binocular alignment prior. It is more constraining than feedforward yoking would be, because reshaping it to compensate an asymmetric plant simultaneously imposes regularization and stability costs that adaptation must trade off against the benefit of corrected alignment. The residual after adaptation saturates is therefore not a failure of plasticity per se but the unique optimum of a multi-criterion objective with no jointly satisfying solution.

The remainder of the paper develops this proposal. Section 2 reviews the Hering picture and its empirical revisions. Section 3 develops the integrator-as-postural-controller view, drawing on the continuous attractor framework. Section 4 formalizes a linearized binocular integrator model, identifies the held-state manifold, and characterizes the achievable set of binocular postures. Section 5 introduces plant asymmetry in the linear regime and derives the residual deviation under a homeostatic learning objective with a stability constraint. Section 6 examines what changes when the plant is genuinely nonlinear, identifies a representational ceiling for any flat-kernel integrator, and connects the linear-versus-nonlinear distinction to the comitant-versus-incomitant clinical dichotomy. Section 7 grounds the abstract priors in brainstem anatomy: sparse and stereotyped connectivity, sign constraints from Dale's law, bilateral symmetry of the wiring, and the dual-pathway architecture for vergence. Section 8 lays out predicted clinical signatures, and Section 9 discusses scope, limitations, and the relationship of the framework to the demonstrably nonlinear control machinery characterized in the Crawford and Tweed literatures.

2. From Hering's law to its empirical revisions

Hering's 1868 proposal that the two eyes are innervated as a single organ shaped a century of oculomotor thought. In its strong form, every binocular eye movement is generated by the sum of a conjugate command, applied identically to corresponding muscles of the two eyes, and a

vergence command, applied antisymmetrically. The neural correlate this implies is a controller in which descending signals split into exactly two channels, distributed by fixed, anatomically determined yoking ratios.

The strong form has been falsified at multiple levels. Zhou and King (1998) showed that PPRF neurons previously assumed to encode conjugate velocity commands actually encode monocular saccadic commands, with separate populations driving each eye. The Mays group has documented saccade-vergence burst neurons whose activity does not factor into independent conjugate and vergence components. Maiello, Harrison, and Bex (2016) used a dichoptic adaptation paradigm to show that the saccadic system can install opposite-direction corrections in the two eyes, demonstrating an adaptive degree of freedom that strict Hering forbids. Schor's phoria adaptation work, and the Maxwell/Schor extensions to context-dependent vergence adaptation, similarly establish that the brain has more adaptive parameters available than yoking allows.

A weaker form of the Hering claim survives: that the brain's controller class includes Hering-like decompositions as natural axes, even if it does not strictly enforce them. The Hybrid Binocular Control framework (Ramat and colleagues) makes this explicit, combining a yoked conjugate subsystem with two slower monocular controllers. The empirical evidence, taken together, supports this softer view. Binocular control is approximately Hering for fast transient movements and increasingly disconjugate for slower postural and adaptive components.

This empirical landscape exposes the question we want to address. If the controller class is richer than strict Hering, then strict Hering is the wrong place to look for the structural prior that limits adaptation. The Hering decomposition is a natural basis but not a hard constraint. So where does the actual constraint live, and why does it produce residual deviations with a characteristic clinical structure?

Our answer: it lives in the integrator. The integrator is what enforces stable held postures, and its connectivity must satisfy stability constraints that the feedforward command pathway does not. The strong form of Hering is approximately what the integrator's natural connectivity geometry produces, and the limits on adaptive correction are limits on how far the integrator's connectivity can be reshaped without compromising the gaze-holding function it exists to perform.

3. The integrator as postural controller

The neural integrator for horizontal eye position resides in the nucleus prepositus hypoglossi and adjacent medial vestibular nucleus; the vertical-torsional integrator is in the interstitial nucleus of Cajal. Functionally, the integrator transforms transient velocity commands into the tonic motoneuron firing rates required to hold the eye against orbital elastic forces. Mathematically, it implements an approximate temporal integration whose long time constant arises from recurrent excitation tuned to produce a near-zero eigenvalue in the dynamics matrix.

The continuous-attractor framework, introduced for oculomotor integration by Seung (1996) and developed by Cannon and Robinson, makes this precise. A network with appropriately tuned recurrent connectivity has a continuous one-parameter family of stable fixed points — a line attractor. Drive along the attractor shifts the network to a new fixed point; perturbations orthogonal to the attractor decay back. The motoneuron readout of the attractor state is the held eye position. Multistable attractors of higher dimension extend the same idea to multiple integrated channels: a 3D attractor manifold supports the held states of a single eye in 3D, a 6D manifold supports binocular held states.

The integrator is therefore not a passive intermediary between command and motoneurons. It is the postural controller, and the geometry of its attractor manifold is the geometry of the achievable set of held postures. Whatever the descending command does transiently, the only states the system can dwell in are those on the manifold.

This recasts the binocular alignment problem. Aligned binocular fixation requires that for every gaze direction, the integrator can settle into a state whose motoneuron readout produces a binocularly aligned eye configuration. The set of aligned binocular held states is a 6D submanifold of the 12D joint motoneuron rate space (one rate per muscle), parameterized by version and vergence target. Whether the integrator can occupy these states depends on whether the integrator's attractor manifold, projected through the plant, contains them. Adaptation can reshape the integrator's connectivity, and therefore the manifold, but only within the limits imposed by the integrator's other functional requirements: stability of held states, low metabolic cost, and accurate integration of velocity commands into position.

The homeostatic learning framework developed in our prior work captures these requirements as a constrained optimization. The recurrent weight matrix is regularized toward minimal recurrent excursion under behaviorally relevant input statistics, while a data fidelity term enforces accurate integration. We propose that binocular alignment fits into this framework as an additional fidelity term, demanding that the manifold's projection through the plant include the binocularly aligned configurations. The natural alignment of the three terms — minimal recurrence, accurate integration, aligned held states — under a symmetric plant produces the empirically observed near-Hering decomposition. Their misalignment under an asymmetric plant produces the residual.

4. A linear theoretical framework

Let $r \in \mathbb{R}^N$ be the activity of integrator neurons projecting to extraocular motoneurons, with N large. The integrator obeys

$$\tau \dot{r} = -r + Wr + B_c c(t)$$

where W is the recurrent connectivity, $c(t)$ is the descending command, and B_c is the input projection. Let $A = W - I$. Held states are zeros of $\dot{r}$, given by the kernel of A when $c \equiv 0$. For

graded held states across a continuous range of fixations, A must have a near-zero subspace of dimension equal to the number of postural degrees of freedom — six for binocular control.

The motoneuron firing rates derive from r by a fixed projection, $m = Mr$, with M a $12 \times N$ matrix. The eye configuration produced by motoneuron rates m is determined by the plant, $\theta = Pm$, with P a 6×12 matrix encoding the muscle pulling-direction Jacobian and tonic length-tension properties. Composing, the held binocular eye configuration achievable from integrator state r is $\theta = PMr$.

The achievable set of held binocular configurations is therefore

$$\mathcal{H} = \{PMr : r \in \ker A\}.$$

The structural prior on binocular alignment is the geometry of $\mathcal{H}$ as determined by the structure of $\ker A$. The dimension of $\mathcal{H}$ is at most $\dim(\ker A)$, and equals it generically when PM has full rank on $\ker A$. We assume this throughout: $\dim \ker A = 6$ and the projection is full rank, so $\mathcal{H}$ is a 6D linear subspace of the 12D motoneuron-rate-image $\mathbb{R}^6$ of plant outputs.

Under symmetric plant geometry — left and right plants related by mirror reflection — the standard Hering decomposition emerges naturally. Decompose $\ker A = K_v \oplus K_d$, where K_v is a 3D conjugate subspace, in which neurons projecting to yoked muscle pairs co-fire, and K_d is a 3D disjunctive subspace, in which they antifire. Under symmetry, PM maps K_v to the 3D conjugate eye configuration subspace and K_d to the 3D vergence subspace, and these two subspaces are exactly the standard binocular alignment basis. Adaptation can independently scale these channels and add tonic offsets — corresponding to phoria adjustments in the language of the clinical literature — within the natural Hering basis without restructuring $\ker A$ at all. This is why concomitant deviations of small magnitude adapt smoothly: they fit inside the homeostatic optimum.

Now suppose the plant is asymmetric: P no longer has the symmetric block structure that aligns the natural Hering kernel with the binocular alignment basis. The set of motoneuron rate patterns producing binocularly aligned eye configurations is now a different 6D subspace $\mathcal{R}^* \subset \mathbb{R}^N$, related to but rotated away from the natural Hering kernel. For perfect alignment, we would need $\ker A = \mathcal{R}^*$. The question becomes whether adaptation can reshape A to install this kernel, and at what cost.

5. Symmetric versus asymmetric plants and the stability tax

Reshaping $\ker A$ to align with $\mathcal{R}^*$ amounts to perturbing the recurrent weight matrix W such that the new dynamics matrix $A' = W' - I$ has its near-zero eigenvalue subspace along $\mathcal{R}^*$ rather than the natural Hering directions. Two costs constrain how far this reshaping can go.

The first is the homeostatic regularization cost. Following our prior framework, the learning objective penalizes recurrent excursion via a term proportional to $\text{tr}(AC A^T)$, where C is the covariance of integrator activity under natural behavior. This term favors weight matrices that are small in directions that are heavily exercised. Reshaping the kernel to include disconjugate-rate directions — directions in which yoked pairs fire unequally — requires nonzero entries in W along those directions. If natural binocular behavior has the disconjugate-rate directions of $\mathcal{R}^*$ at low variance under C , then installing them in $\ker A$ requires the regularizer to absorb a cost it would prefer not to pay.

The second is the stability cost. The integrator's gaze-holding function requires that the eigenvalues along $\ker A$ be precisely zero. Generic perturbations to W that move the kernel direction also tend to push some eigenvalues slightly negative (producing centripetal drift toward primary position, the leak that underlies gaze-evoked nystagmus) or, less commonly, slightly positive (producing centrifugal runaway). The clinically dominant failure mode is the leaky case, because the homeostatic regularizer tends to push eigenvalues toward -1 rather than 0^+ — smaller weight magnitudes generically produce sub-unity recurrent gain. This gives the stability tax a specific empirical signature: residual misalignment couples preferentially with centripetal drift and gaze-evoked nystagmus rather than with runaway instability. Maintaining zero eigenvalues exactly along a non-natural kernel requires fine-tuning of W in directions that the homeostatic regularizer pushes toward zero. The set of weight matrices with both the right kernel direction and exactly-zero eigenvalues along it is a lower-dimensional submanifold of weight space, and adaptation operating on a regularized objective will not generically converge to it.

The full adaptive optimum is the solution of a constrained minimization,

$$\min_W \alpha \mathcal{E}_{\text{align}}(W; P) + \beta \mathcal{E}_{\text{integrate}}(W) + \gamma \text{tr}(WCW^T) \quad \text{s.t. stability of } \ker(W - I),$$

where $\mathcal{E}_{\text{align}}$ measures binocular misalignment averaged over fixations, $\mathcal{E}_{\text{integrate}}$ measures integration accuracy, and the trace term is the homeostatic regularizer. Under a symmetric plant, all four terms are jointly minimized at the same W : the natural Hering integrator. Under an asymmetric plant, no single W jointly minimizes them, and the optimum is a compromise.

The structure of this compromise determines what the residual looks like. Linearizing around the symmetric optimum and treating plant asymmetry as a small perturbation δP , the residual misalignment scales linearly with δP , while the additional homeostatic and stability costs scale quadratically with the weight perturbation δW that the system installs. The optimum sets δW proportional to δP with a coefficient determined by the regularization weights. Critically, the residual misalignment and the residual stability cost are coupled: reducing one increases the other. So the predicted clinical fingerprint of an asymmetric plant under chronic adaptation is not pure misalignment, and not pure gaze-holding pathology, but a specific linear combination of the two whose ratio is a property of the regularization landscape.

This is the stability tax. The brain cannot install the correct manifold geometry for free. Every degree of disconjugate-rate kernel structure carries a near-zero-eigenvalue perturbation that manifests as subclinical drift. The empirically observed coexistence of strabismus with subtle integrator-like signs — gaze-evoked nystagmus, post-saccadic drift, rebound nystagmus — is a direct prediction of this framework, not a coincidence to be explained by independent pathologies.

6. Linear versus nonlinear asymmetries: the representational ceiling

The analysis of Section 5 treats plant asymmetry as a small linear perturbation: P_L and P_R are constant matrices that differ from each other, and the alignment-correct motoneuron rate manifold $\mathcal{R}^*$ is a flat 6D linear subspace rotated away from the natural Hering kernel. In this regime, reshaping $\ker A$ to track $\mathcal{R}^*$ is a one-shot weight perturbation problem. The cost is paid in regularization and stability, but once paid, the compensation holds at every fixation, because a linear plant produces the same alignment error at every operating point given the same residual kernel mismatch. The clinical correlate is a deviation of constant magnitude across all gaze directions — concomitant strabismus, in standard terminology.

Plant nonlinearities break this picture in a categorically distinct way, and the simplest nonlinearity makes the distinction sharp. Consider a saturation: a muscle whose force-rate curve plateaus above some firing rate. The local plant Jacobian then depends on operating point. Near primary position, with all muscles in their linear range, the plant is approximately a constant P_0 . At eccentric gaze toward a saturating muscle's field of action, the row of $P(\theta)$ corresponding to that muscle has reduced effective gain. The motoneuron rate pattern that produces an aligned binocular configuration is therefore not constant: it depends on where the eyes are pointing. The alignment-correct manifold $\mathcal{R}^*$ is no longer a flat subspace of motoneuron rate space but a curved submanifold whose local tangent direction varies smoothly with operating point.

A linear integrator with a flat 6D attractor manifold cannot exactly contain a curved $\mathcal{R}^*$, no matter how its kernel is rotated. The best a flat kernel can do is be tangent to the curved manifold at a single operating point — typically primary position, where adaptation is most strongly driven. Everywhere else, the flat kernel and the curved $\mathcal{R}^*$ diverge, and the divergence scales with the local curvature of $\mathcal{R}^*$. Even with unlimited freedom on W , alignment is fully corrected only at one fixation; the residual at other fixations is a representational ceiling of the linear-integrator architecture, not a failure of the learning rule. This is incomitant strabismus, and the incompletely-adapted character of paralytic and restrictive deviations follows from the architecture rather than from any defect of plasticity.

The interaction of saturation with Hering yoking is specifically diagnostic. Yoked drive forces equal command to a paretic muscle and its contralateral yoke partner. If the paretic muscle saturates earlier than its partner, then in its field of action increasing yoked drive does nothing for the paretic eye while continuing to drive the partner, which has not yet saturated. The clinical sign of overaction of the contralateral yoke in paralytic strabismus is precisely this — yoking plus

saturation, with the asymmetry concentrated where the saturation bites. The integrator's flat kernel cannot route around the saturation, because the kernel structure is fixed in motoneuron rate space, while the saturation is a property of the rate-to-force map and lives downstream of the integrator entirely.

Compensation strategies for nonlinear asymmetries are accordingly different from the linear case. A flat integrator kernel can be rotated, but it cannot be curved. Curvature requires either gaze-dependent modulation of the integrator's effective dynamics — context inputs that scale or shift the recurrent weights as a function of fixation, the cerebellar gain-field mechanism — or a feedforward inverse model that pre-distorts commands to anticipate the nonlinear plant. Both exist biologically. Context-dependent VOR adaptation, in which gain is modulated as a function of head orientation or vergence state, has been documented extensively (Shelhamer; the McLaughlin saccade-adaptation lineage; Maxwell & Schor for vergence). Inverse modeling of the half-angle rule and Listing's constraint has been argued for repeatedly in the Crawford and Tweed literatures. But both mechanisms add representational complexity beyond the flat integrator, and both have their own learnability constraints — they require richer error signals (gaze-conditional rather than gaze-averaged) and recruit cerebellar circuitry rather than brainstem plasticity alone.

Spread of comitance receives a particularly clean explanation in this framing. It is the system finding the best flat-kernel approximation to a curved $\mathcal{R}^*$. The flat approximation is uniformly mismatched across gaze directions, which is concomitant in clinical language; the alternative — installing gain-field machinery to track the curvature — is more expensive in adaptive resources. The system prefers the cheaper flat approximation when the cost of the gain-field solution exceeds the costs of the residual misalignment plus suppression and sensory adaptation. Spread of comitance is therefore not a failure of the cerebellar gain-field mechanism per se, but the optimum of a cost-benefit calculation that often goes against committing the gain-field machinery to a problem the flat-kernel approximation can partially solve.

The taxonomy that emerges from this section: comitant deviations sit inside the linear analysis of Section 5 and are governed by the Hering-kernel rotation problem and its stability tax. Incomitant deviations exceed the representational capacity of the flat kernel and require gaze-dependent compensation that may or may not be deployed. Spread of comitance is the boundary phenomenon — the system actively choosing the flat-kernel approximation over committing to nonlinear compensation. Each clinical category corresponds to a specific level in the hierarchy of representational capacity available to adaptation.

7. Anatomical implementation: how the integrator architecture instantiates the prior

The structural prior characterized abstractly as a constraint on the recurrent weight matrix W corresponds to specific facts about the brainstem architecture that implements the integrator. These facts make certain kernel directions cheap and others expensive in concrete biological

terms — the prior is not an abstract regularization preference but a consequence of which connections exist, which neurons are excitatory or inhibitory, and what bilateral symmetries are baked into the substrate.

Horizontal gaze holding is implemented by a bilaterally distributed circuit centered on the nucleus prepositus hypoglossi and adjacent medial vestibular nucleus. Each side hosts populations of excitatory burst-tonic and tonic neurons whose recurrent excitation produces the long integrator time constant, together with inhibitory commissural neurons that couple the two sides through GABAergic projections across the midline. The output projects to the ipsilateral abducens nucleus, which contains both motoneurons innervating the ipsilateral lateral rectus and internuclear neurons whose axons cross the midline and ascend in the medial longitudinal fasciculus to terminate on the contralateral medial rectus subdivision of the oculomotor nucleus. Activation of one side's NPH thus drives both the ipsilateral abducens and, via the abducens internuclear pathway, the contralateral medial rectus — producing conjugate horizontal gaze through a hardwired projection pattern. The vertical and torsional integrator in the interstitial nucleus of Cajal has a structurally analogous but distinct organization, with its own commissural couplings and projections to the trochlear and oculomotor nuclei.

Several features of this architecture function as structural priors on the achievable kernel of W .

First, the sparsity of the connectivity matters. The integrator's recurrent weight matrix is not a dense $N \times N$ generic matrix but a structured sparse matrix in which most pairs of neurons do not synapse on each other at all. The dimensions of W that synaptic plasticity can adjust are precisely those corresponding to existing axonal projections, which themselves were laid down during development. Reshaping the kernel of W in directions that would require nonexistent projections is impossible without anatomical rewiring on developmental timescales. The set of accessible W matrices is therefore not a full vector space but a low-dimensional submanifold determined by the pre-existing wiring diagram, and the abstract regularization landscape of Section 5 should be understood as restricted to this submanifold throughout.

Second, sign constraints from Dale's law. Each integrator neuron is either excitatory or inhibitory, and the sign of its outgoing weights is fixed by its neurotransmitter identity. The rows of W corresponding to glutamatergic neurons are non-negative; those corresponding to GABAergic commissural neurons are non-positive. Plasticity adjusts magnitudes within these sign constraints. Many disconjugate-rate kernel directions that would be straightforwardly accessible to an unconstrained W are not accessible to a sign-constrained W — they would require recruiting opposite-sign drive from neurons whose neurotransmitter identity forbids it. The system can sometimes route around this through intermediate inhibitory populations, but each such detour adds latency, recruits additional metabolic cost, and introduces additional eigenvalues whose stability must be separately maintained.

Third, bilateral symmetry of the wiring. The same anatomical pattern recurs on the two sides of the brainstem with mirror symmetry. This induces a corresponding symmetry on W : under

reflection of the neural population across the midline, W is approximately invariant. Symmetric matrices have symmetric and antisymmetric eigenvectors, which decompose naturally into conjugate (left-right symmetric) and disjunctive (left-right antisymmetric) modes. The Hering decomposition of the kernel is therefore not a learned property but a direct consequence of the bilateral symmetry of the underlying anatomy. The two channels are easy to maintain because they are eigenvectors of the symmetry operation. Non-Hering kernel directions are not eigenvectors, and require breaking the symmetry of W — which costs both regularization (asymmetric components are penalized by $\text{tr}(WCW^T)$ when C inherits the bilateral symmetry of natural binocular behavior) and stability (asymmetric perturbations split eigenvalue degeneracies and tend to produce nonzero real parts in modes that should remain near zero).

Fourth, a dual-pathway architecture for vergence that exists only along certain axes. Medial rectus motoneurons receive input both from the abducens internuclear pathway, which carries conjugate signals, and from a separate input arising in the supraoculomotor area, which carries vergence-specific drive. The vergence pathway is parallel to and partially independent of the conjugate pathway, and is the anatomical substrate for horizontal disjunctive control. Crucially, this dual-pathway architecture is well developed only for horizontal vergence; vertical and torsional disjunctive channels lack equivalent dedicated inputs. This predicts the empirical observation that vertical phorias are smaller in magnitude than horizontal but more refractory to adaptation, and that torsional alignment adapts poorly. The disjunctive channel for these axes does not have the anatomical substrate to support strong adaptive correction, regardless of how rich the error signal might be. The structural prior is, in these directions, near-impermeable.

Together, these architectural features convert the abstract regularization-and-stability trade-off of Section 5 into a concrete biological prior. The cheapest kernel directions are those consistent with bilateral symmetry, sign-constraint, and existing axonal projections; the natural Hering decomposition occupies precisely these directions. Asymmetric plants demand kernel directions that break one or more of these constraints, and the cost of installing them is paid in symmetry-breaking weights, dependence on weakly developed pathways with low intrinsic gain, or recruitment of indirect circuits that introduce their own stability problems. The stability tax has, in this view, a specific anatomical interpretation: it is the cost of operating the integrator with a W matrix that is mismatched to the bilaterally symmetric, sign-constrained, sparsely connected substrate the brainstem provides. The plasticity that does occur — synaptic gain changes within existing pathways, slow drifts in commissural balance, adjustments to supraoculomotor vergence drive — operates within this substrate rather than reshaping it.

8. Predictions and clinical correspondence

Several otherwise-puzzling clinical observations follow naturally from the framework.

Spread of comitance — the gradual conversion of an initially incomitant deviation into a comitant one over months — is, in this view, not a failure of adaptation but the homeostatic regularizer's preferred solution. Faced with an asymmetric plant, the system can either reshape

the kernel to compensate (high homeostatic cost, residual stability tax) or shift the entire manifold so that all gaze directions are equally affected (low homeostatic cost because the kernel structure is preserved, residual is uniform misalignment). The latter is energetically preferred and is what is observed clinically.

The developmental sensitivity of binocular alignment follows from the plasticity profile of the integrator. The integrator's basic kernel structure is set during early development, when the relevant behavioral statistics that shape C are still being established. Plant asymmetries present during this period can be absorbed into the natural kernel as it forms; asymmetries arising after the kernel consolidates require perturbation of an already-stabilized structure and pay the full stability tax. This predicts that infantile strabismus and adult-onset palsies should produce qualitatively different residuals: developmental cases consolidate into kernels that are intrinsically miscalibrated but stable, while acquired cases install perturbed kernels that always carry residual stability costs. Existing clinical observations of stronger gaze-holding abnormalities in acquired than developmental strabismus are consistent with this distinction, though the comparison has not been quantified systematically.

Disconjugate adaptation paradigms (Maiello et al. 2016; Bucci & Kapoula 1997) probe the system's ability to install non-Hering structure on a short timescale. Our framework predicts that such adaptation is possible because the kernel of A is locally plastic, but that the magnitude of achievable disconjugate adaptation is bounded by the regularization weights. Strabismic subjects, whose kernels have already been perturbed chronically, should show reduced range of additional disconjugate adaptation, consistent with Bucci and Kapoula's findings.

Vertical and torsional alignment should be more refractory to adaptation than horizontal. This follows because the relevant adaptive error signals (vertical and cyclo-disparity) are sparser and weaker than horizontal disparity, so the data fidelity term has less leverage relative to the regularizer. Clinically, vertical phorias are smaller in magnitude but harder to fully resolve, and torsional alignment is famously poorly adapted — both consistent with the framework.

The framework also predicts a class of experiments not yet performed. If adaptation is forced via prism or post-surgical realignment, the recovered alignment should come bundled with measurable integrator-level changes — small drifts, time-constant shifts, altered gaze-holding statistics. The signature should be predictable from the magnitude of the imposed alignment shift. To our knowledge no study has measured high-precision integrator function before and after surgical alignment in adult strabismus; this would be a direct test.

9. Discussion

We have argued that the structural prior limiting adaptive compensation of binocular misalignment lives in the recurrent connectivity of the neural integrator, not in any feedforward yoking rule. The integrator's continuous-attractor architecture imposes a manifold of held postural states whose geometry is set by the joint requirements of stability, integration accuracy, and homeostatic minimal-recurrence. Under symmetric plants this manifold's natural geometry

is approximately Hering, which is why the Hering description has explanatory force. Under asymmetric plants the alignment-correct manifold differs from the natural one, and adaptation cannot install it without paying coupled regularization and stability costs. The residual deviation has a characteristic joint signature in alignment and gaze-holding that distinguishes it from peripheral plant abnormalities.

The framing has two virtues over the strict-Hering account. First, it is consistent with the empirical evidence that the brain has more than two adaptive channels, because the constraint is on the integrator's manifold geometry rather than on the feedforward command pathway. Second, it makes coupled predictions about alignment and gaze-holding that the feedforward account does not, opening clear empirical tests. The taxonomy of Section 6 further unifies several otherwise-disconnected clinical categories under a single representational hierarchy.

A natural objection to the framework is that the brain is not strictly linear, and a strictly linear analysis of plant asymmetries therefore misses much of what the system actually does. The objection has force, and the work of Crawford and colleagues makes it concrete. Klier, Wang, and Crawford (2001) showed that while the superior colliculus encodes 2D saccade commands, downstream brainstem circuitry implements the 3D rotational kinematics required by Listing's law and the half-angle rule, performing a position-dependent nonlinear transformation that no purely linear feedforward stage could produce. Smith and Crawford, and Klier and Crawford in subsequent work, have mapped the brainstem implementation of these 3D commands in detail, establishing that the oculomotor controller performs genuine nonlinear computation in service of 3D rotational geometry. The brain is demonstrably not linear; it has machinery for nonlinear control.

Two distinctions, however, prevent this from undermining our analysis. First, the nonlinearities documented in the Crawford literature are *structural* rather than *adaptive*. They implement the geometric necessities of rotation in three dimensions — Listing's plane, the half-angle rule — and are stereotyped, robust to perturbation, and only minimally plastic. They do not appear to be the substrate by which the brain compensates for unpredictable plant changes such as muscle weakness, pulley displacement, or scleral aging. The adaptive plasticity that does respond to such changes — phoria adaptation, monocular saccade adaptation, vergence-state-dependent gain adjustment — has been characterized empirically as approximately linear over the operating range tested, with state-dependence as the principal mechanism for nonlinear correction. The Maxwell-Schor context-dependent vergence adaptation work and the Shelhamer state-conditional VOR adaptation lineage both establish that gaze- and context-dependent gain modulation is available, but they also establish its limited dimensionality and slow learning rate compared to the linear gain-and-bias channels of the basic integrator.

Second, a substantial fraction of the apparent nonlinear compensation in the oculomotor system is implemented passively, in the plant rather than the controller. Demer's pulley work establishes that the connective-tissue pulleys constraining each rectus muscle enforce the half-angle rule mechanically — the muscle's pulling direction shifts with eye position by half the gaze angle as a

passive consequence of the pulley geometry, without any neural computation. The brain does not need to issue a position-dependent torsional command to satisfy Listing's law because the pulleys do most of the relevant geometry. This further sharpens the distinction: where the plant provides passive nonlinear structure, the neural controller can be approximately linear and still produce empirically nonlinear behavior. Where the plant nonlinearity is *not* automatically compensated by passive structure — as in our saturation example, in pulley heterotopias that disrupt the half-angle rule, or in muscle paresis where the rate-to-force map shifts unpredictably — the neural controller bears the burden, and the linear-integrator ceiling of Section 6 becomes operative.

The picture that emerges is a layered architecture: passive plant nonlinearities (pulleys, length-tension curves) handle the geometric necessities; structural neural nonlinearities (Listing's machinery, half-angle commands) handle the residual geometric corrections; an approximately linear adaptive integrator handles plant calibration; and slow cerebellar context-dependent modulation handles whatever nonlinear residual the linear integrator cannot capture. Each layer has its own representational capacity and its own plasticity timescale. Comitant deviations are absorbed at the integrator level. Incomitant deviations exceed integrator capacity and demand cerebellar gain-field engagement, which may or may not happen depending on the cost-benefit calculation discussed in Section 6. Severe pulley pathology disrupts the lowest layer, where no neural compensation is straightforwardly available.

Several limitations of the present treatment should be flagged. The linearized analysis of Section 5 is valid only for small plant perturbations; severe pulley dislocations or muscle palsies move the system into the regime of Section 6 where the flat-kernel approximation is qualitatively inadequate. Listing's law and the half-angle rule introduce additional gaze-dependent constraints that interact nontrivially with the integrator's attractor structure, particularly for vertical and torsional alignment, and we have not analyzed these interactions in detail. The relationship between motoneuron rate patterns and integrator activity, collapsed here into the projection M , is in reality an active read-out involving brainstem premotor nuclei whose own dynamics may further constrain the achievable set. The behavioral covariance C that drives the homeostatic regularizer is itself shaped by the available integrator structure, producing a self-referential loop whose fixed-point analysis we have not undertaken. And the cerebellar gain-field mechanism invoked at multiple points in the argument has been treated as a general capacity rather than as a specific computational architecture; a satisfactory account would require a parallel theoretical treatment of cerebellar plasticity in the same framework.

Despite these limitations, the central proposal stands as a tractable theoretical claim with concrete predictions: the locus of structural priors on binocular alignment is the integrator, the predicted residual signature is coupled misalignment and gaze-holding pathology, comitant and incomitant deviations correspond to distinct levels of representational capacity in the adaptive hierarchy, and the developmental versus acquired distinction follows from when in the integrator's plasticity timeline the asymmetry arrived. Whether this framework explains a

substantial fraction of the variance in clinical strabismus, or only a particular slice of it, is an empirical question that the predictions in Section 8 are designed to address.

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(To be completed; key citations to integrate)

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